HEALTHCONNECT RWANDA

Final Business Insights & Executive Recommendations

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1. Executive Summary

The **HealthConnect Rwanda Data Analytics Project** delivers an integrated Business Intelligence solution designed to improve visibility into healthcare operations, patient demographics, clinical activity, financial performance, resource management, and patient experience across 20 healthcare facilities.

The project involved rigorous data profiling, transformation, relational modeling, DAX measure development, and the deployment of a 6-page Power BI executive dashboard consolidating patient records, appointments, billing, claims, diagnoses, admissions, and facility equipment.

> **Core Finding:** A critical disparity exists between Gross Billed Revenue (**RWF 9.31bn**) and Department-Allocated Revenue (**RWF 771.13M**). Approximately **RWF 8.54bn** remains unallocated due to missing or unmatched appointment references in transactional data, highlighting a key area for operational data governance.

2. Key Financial & Operational KPIs

GROSS BILLED	UNALLOCATED REVENUE	ALLOCATED REVENUE	CLAIM APPROVAL RATE
RWF 9.31B	**RWF 8.54B**	**RWF 771.1M**	**49.98%**
Total revenue billed	Unmatched appointments	Matched to departments	Insurance claims approved

3. Key Business Insights

3.1 Financial Performance & Revenue Accountability

The reconciliation established that total billing revenue consists of **RWF 771.13M** successfully matched to appointments and **RWF 8.54B** that could not be matched to available appointment records. This unallocated amount does not indicate lost revenue, but rather a data attribution and logging problem occurring at intake and billing.

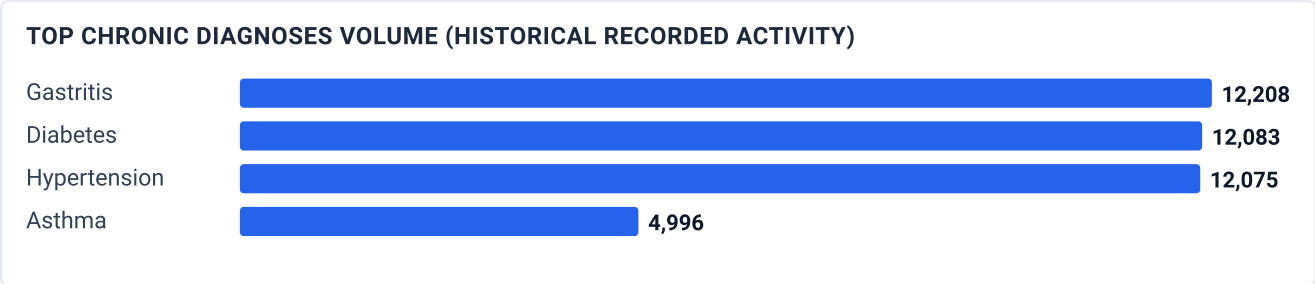


Payment Channels: Collections are evenly distributed across Mobile Money (25.2%), Card (25.1%), Insurance (25.0%), and Cash (24.7%), reflecting high digital adoption among patients.

3. Key Business Insights (Continued)

3.2 Patient Demographics & Chronic Condition Demand

The network covers **20,000 registered patients** across 20 facilities with a balanced gender distribution (50.45% Female, 49.55% Male). Clinical analysis reveals high historical diagnostic activity around four core chronic conditions:



3.3 Insurance Claim & Revenue Cycle Efficiency

The system recorded an overall **Insurance Claim Approval Rate of 49.98%**. The remaining ~50% represents pending, delayed, or rejected claims requiring operational review into documentation requirements and submission protocols across providers (MMI, RSSB, Radiant, Private).

3.4 Operational Resource Constraints

- **Equipment Availability:** 4,000 registered assets across 20 facilities. Recorded as 100% baseline by assumption due to lack of IoT/maintenance logs.
- **Bed & Room Utilization:** All 2,000 beds are marked "Available" in the database baseline. Transactional occupancy logs are currently unavailable.

4. Executive Recommendations

Rec 1: Strengthen Financial Data Integrity & Billing Validation

Mandate `AppointmentID` validation prior to saving billing transactions. Conduct a targeted reconciliation for the 55,000 unlinked billing records to eliminate the RWF 8.54bn unallocated revenue blind spot.

Rec 2: Improve Insurance Claim Management & Workflows

Establish a dedicated revenue-cycle task force to track pending vs. rejected claims, standardize documentation, and shorten reimbursement turnarounds across all key insurance providers.

Rec 3: Scale Outpatient Chronic Care & Pharmaceutical Planning

Align pharmaceutical inventory (e.g., Amlodipine, Metformin, Omeprazole) and outpatient staffing with the high demand observed in Gastritis, Diabetes, and Hypertension care pathways.

Rec 4: Introduce Real-Time Bed & Equipment Operational Tracking

Deploy operational logging for bed admissions/discharges and an equipment maintenance register to calculate true occupancy and downtime metrics across facilities.

5. Management Priority Matrix

PRIORITY	MANAGEMENT ACTION	BUSINESS IMPORTANCE
CRITICAL	Resolve unmatched billing and appointment references	Financial Accountability & Revenue Traceability
CRITICAL	Improve insurance claim processing and status tracking	Revenue-Cycle Efficiency & Working Capital
HIGH	Introduce real-time bed and room occupancy tracking	Capacity Management & Inpatient Flow
HIGH	Implement equipment maintenance monitoring	Operational Continuity & Asset Performance
HIGH	Strengthen chronic disease outpatient management	Clinical Planning & Resource Allocation
HIGH	Improve patient feedback collection & appointment linking	Service Quality & Patient Experience
MEDIUM	Improve admission and 30-day readmission data capture	Clinical Outcomes Evaluation
ONGOING	Establish network-wide data governance framework	Reporting Reliability & Analytics Scale

6. Data Limitations & Metric Interpretation Guide

To ensure accurate board reporting, metrics classified as **Data Constrained (N/A)** must not be interpreted as zero performance, but as areas requiring upgraded operational logging:

METRIC	CURRENT STATUS	ROOT CAUSE / LIMITATION
Bed Utilization Rate	N/A – Data Constrained	All 2,000 beds classified as "Available"; no occupied bed logs.
Room Utilization Rate	N/A – Data Constrained	No room status field or occupied-bed mapping in current schema.
Equipment Availability	100% Baseline Assumption	4,000 registered assets; maintenance & downtime logs unavailable.
Admission Rate	N/A – Data Constrained	Lack of standardized outpatient encounter denominator population.
30-Day Readmission Rate	N/A – Data Constrained	Missing confirmed readmission flag and exact discharge timestamps.
Unallocated Revenue	RWF 8.54 Billion	55,000 billing records lack matching AppointmentIDs in source data.

7. Conclusion & Final Management Message

The HealthConnect Rwanda Data Analytics Project establishes a solid foundation for evidence-based healthcare management across 20 facilities. The 6-page Power BI dashboard delivers immediate executive visibility while clearly illuminating the necessary operational data governance priorities.

> **Final Message to Leadership:** *"The true value of a healthcare analytics platform lies not only in the KPIs it reports, but in its power to expose data gaps and guide strategic action. By resolving transaction linkages and deploying real-time status tracking, HealthConnect Rwanda will transition from descriptive reporting to predictive, proactive healthcare excellence."*